

Errata for the book *SDEs for Science and Engineering*

Uffe Høgsbro Thygesen, October 1, 2026

Note: This errata concerns the printed version (2023). I would be grateful if you report further errors by e-mail to uhth@dtu.dk.

Errors

Page	Position	Correction
49	l. 13	Replace $p > 0$ with $p \geq 1$.
57	Theorem 3.9.5, 1.	Replace $p \geq q > 0$ with $p \geq q \geq 1$
57	Theorem 3.9.5, 2.	Replace $p > 0$ with $p \geq 1$
58	l. 7	Replace $p > 0$ with $p \geq 1$.
87	Exercise 4.17, q. 2	Replace $a_j = P_{ij}/P_{ii}$ with $a_j = -P_{ij}/P_{ii}$.
87	Exercise 4.18	Replace $M_i = \sum_{j=1}^i X_j$ with $M_i = \sum_{j=1}^i X_j$
113	Exercise 5.8, q. 2 and 3	Note that a more compact “Hamiltonian” method exists
121	l. 4	Replace the potential with $u(x) = -1/2rx^2 + 1/4qx^4$
174	l. 2	Replace $\mathbf{E} X_t ^2 \leq (x_0^2 \cdots$ with $\mathbf{E} X_t ^2 \leq (x_0 ^2 \cdots$
194	l. 3	Replace $\{B_u : 0 \leq u \leq u\}$ with $\{B_u : 0 \leq u \leq s\}$
197	l. 5	”The white curve” should be ”The black curve”
219	Last displayed formula	Replace $\mathbb{Q}\{Z \mathcal{G}\} = \mathbb{E}\{L_T Z \mathcal{G}\}$ with $\mathbb{Q}\{Z \mathcal{G}\} = \mathbb{E}\{L_T Z \mathcal{G}\}/\mathbb{E}\{L_T \mathcal{G}\}$
219	l. 5 from bottom	Replace $\mathbb{Q}\{W_t \mathcal{F}_s\} = \mathbb{E}\{W_t L_t \mathcal{F}_s\}$ with $\mathbb{Q}\{W_t \mathcal{F}_s\} = \mathbb{E}\{W_t L_t \mathcal{F}_s\}/L_s$
227	Exer. 9.8, q. 2	Replace X_t with Y_t (two places)
227	Exer. 9.8, q. 3	Replace $e^{(-2cy+\nu)/2}$ with $e^{-(2cy+\nu)/2}$. Add $n = 4\lambda\xi/\gamma^2$.
250	after “for the variance”	Replace $-K_t C^\top$ with $-K_t C$.
250	after “and”	Replace $-K_t C^\top$ with $-K_t C$.
254	Second line from bottom	Replace $\log f$ with $-\log f$
280	l. 8	In “... solution v to such ...”, replace v with h
280	6th displayed formula	Replace v with h
308	l. 2	Replace $\max\{Y_t : t \geq 0\}$ with $\max\{\log X_t : t \geq 0\}$
326	Bottom line	Replace h with h^{μ^*}

Spelling errors etc.

Page	Position	Correction
(Multiple	Places)	Change “In stead” to instead
24	Footnote	Change ”Physcially” to ”Physically”
55	l. 10	Change “Is it easy” to “It is easy”
86	l. 6	Replace “Markov process” with “process with independent increments”
86	l. 7	process “with” $N_0 = 0$
87	last line of text	... discretized total variation “of Brownian motion”
90	Fig. 5.1 caption	Replace “drive by” with “driven by”
99	l.2	Replace “autocorrelation” with “autocovariance”
103	After (5.15)	Replace “autocorrelation” with “autocovariance”
104	Fig. 5.6	Replace “autocorrelation” with “autocovariance”
105	(5.19)	This is an approximation. See modified text.
107	Last sentence before 5.10	This “is” at ...
111	Second sentence	Replace “drive” with “driven”
113	Ex. 5.8	Replace the Kronecker argument with the Hamiltonian (also in ...)
114	Ex. 5.9	Replace “Ito equation” with linear differential equation driven by ... Add hint to deal with the timevarying coefficients
124	Thm 6.3.2, item 2	Brackets around the integrand
124	Ito bio	... held positions “at” Princeton ...
125	Thm 6.3.2. item 4	Define It in item 4, not in item 5
141	l. 10	$Y_t = h(X_t)$: add “where h is smooth”
143	Bottom	Replace “in stead” with “instead”
145	l. 3	Replace $X_{t_i}]$ with X_{t_i}
146	l. 1 of 7.4	Replace “of and” with “of any”
174	l. 9	Add brackets around the integrand for $\mathbf{E}S_t$
188	After def 8.6.1	Replace “for all V ” with “for all h ”.
194	Last paragraph in 9.2	Replace “when then” with “when the”
205	3rd displayed formula	Add brackets around integrand
205	3rd eq. for $\dot{\Sigma}(t)$	Add brackets around integrand
206	l. 13	“ ... expect stationary distribution ...”: Add plural “tionS”
207	Ex. 9.8.2	Change “Iogistic” to “Logistic”
210	Bottom derivation	Add brackets around integrands (2 places)
212	L. 9	Replace “we could also shown” with “... show”
212	L. 9 from bottom	Change “important: it” to “important: It”
214	l. 11	Change “Gaussian distributed” to “Gaussian distribution”
215	l. 10	... the support “of” the distribution ...
218	Ex. 9.11.1	Change “The Brok wnian” to “the Brownian”
229	Exercise 9.15	Replace ”of in” with ”of”
252	Last sentence before 10.7.3	Replace “contains” with “contain”
252	Mid-page	Replace “autocorrelation” with “autocovariance”
272	After (11.5)	Add full stop
275	theorem 11.6.1	Replace ”is C^2 satisfies with ”satisfies”
285	After 1rst eq.	Add full stop
292	l. 9 from bottom	... it is “a” robust ...
292	l. 2 from bottom	... also ask “if” non-eq ...
294	l. 4	Replace “satisfies” with “satisfy”
295	(12.6)	Add IC $S_0 = 1$
298	l. 1	3 Replace “and that $x > 0$ ” with “and that the initial condition is $x > 0$ ”
306	l. 15	Replace ”There, we found that a” with ”There, we found a”
306	l 3 from bottom	Replace EX_t with $\mathbb{E}X_t$.
315	Middle of page	Next, if there exists “a” Lyapunov ...
327	Mid-page	Add bracket around integrand.
329	(13.10)	Make notation consistent with previous section
329	(13.10)	Add brackets around integrand
358	Last paragraph	Replace “policuiteration”.
358	Last paragraph	Include exercise about numerical solution of Riccati equation